

SPLATXTRACT: TRACTABLE GAUSSIAN SPLATTING VIA OPEN-WORLD REGION-OF-INTEREST EXTRACTION AND REFINEMENT

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Abstract

We present a task-conditioned refinement for Gaussian Splatting (GS) that enables robots or human operators to selectively extract region-of-interest (ROI). Given a GS map, SplatXtract supports ROI refinement, preserving map consistency while meeting close to real-time constraints required for interactive perception. SplatXtract decouples semantic ROIs from the GS map, allowing flexibility. integration with external and novel perception models.

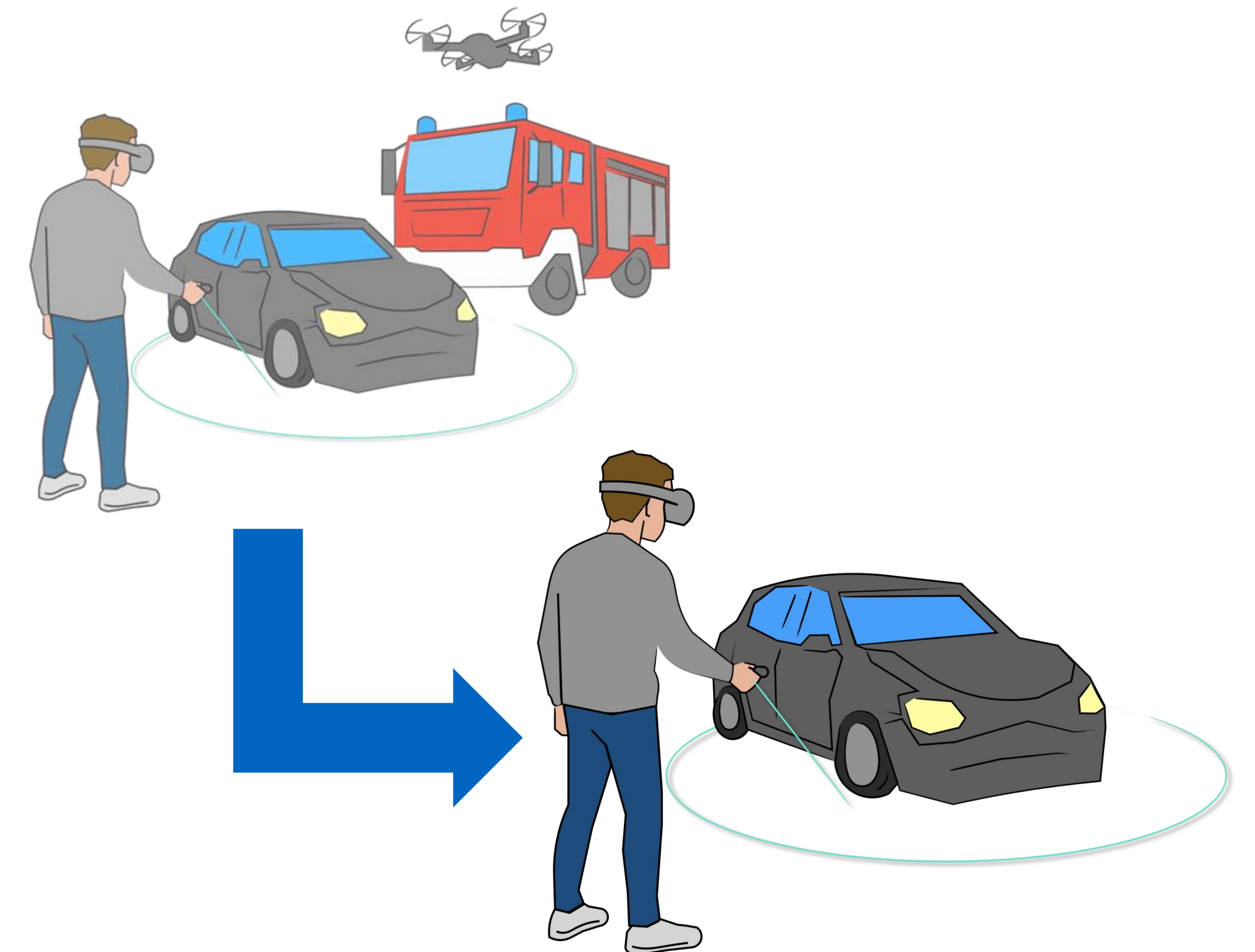
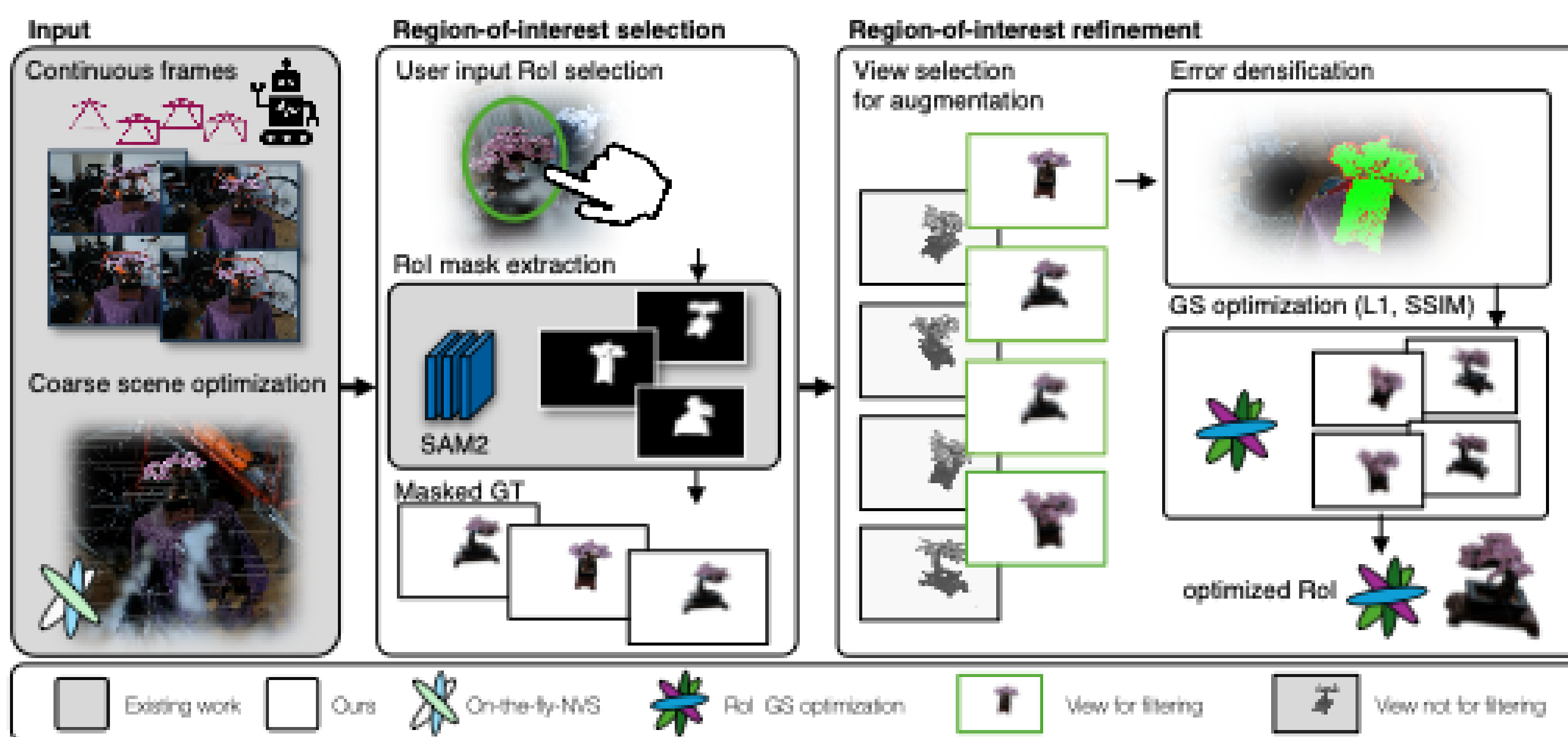
Introduction

Problem. Existing semantic Gaussian Splatting (GS) approaches embed features during training, requiring preprocessing and fixed class labels. This limits flexibility and prevents close to real-time interaction needed for VR teleconsultation and robotic perception.

Contributions.

- User-guided open-world ROI extraction → no fixed class labels.
- Novel error-mask densification to improve reconstruction within the ROI.
- SfM-free pipeline built on on-the-fly NVS → bounded latency for human-in-the-loop use.
- End-to-end VR integration for interactive exploration and ROI annotation.

Method



1. Coarse scene

On-the-fly NVS builds a coarse GS map with camera poses (10–20 FPS, 480×640).

2. ROI selection

User draws ROI in VR; SAM2 propagates a binary ROI mask across all keyframes.

3. Coverage filtering (CGS)

Greedy view selection (4 iters) maximises uncovered Gaussian coverage under ROI masks.

4. Error-mask refinement

Densify high-error regions; 3k iters with masked L1+SSIM loss.

Evaluation

+ 14.039 in PSNR

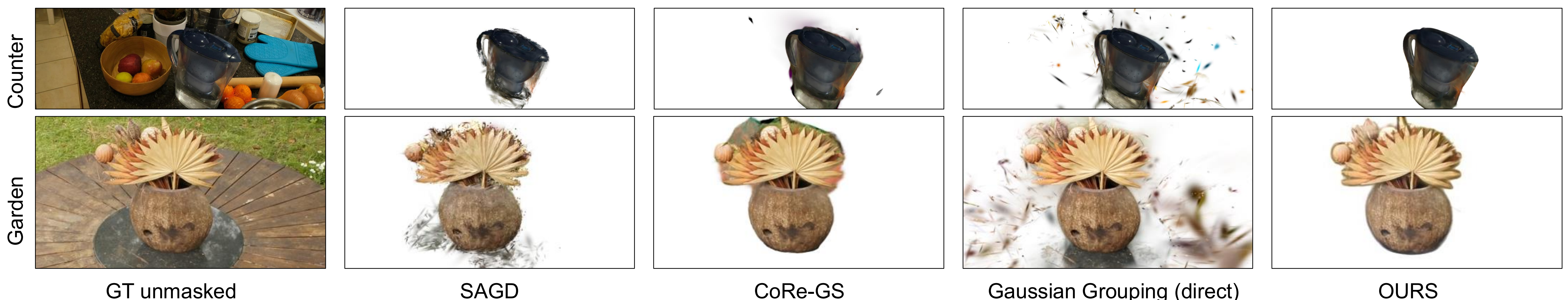
compared to voting-based object removal on TUM RGB-D

+ 6.374 in PSNR

compared to the best performing approach (CoRe-GS) on MipNeRF360

89.7 in mIoU

2. best on 3D-OVS in segmentation quality using ROI strategy



Discussion, Future Work and Conclusion

- Semantic embedding is **not necessary** for effective object-level refinement, mask-guided filtering and error-driven densification suffices.
- Open-world ROI selection enables **flexible post-processing**: drone search-and-rescue, mobile medical teleconsultation, robotic navigation targets.
- **Limitations**: assumes continuous frames with smooth motion; ROI fidelity depends on SAM2 mask quality in cluttered scenes.
- **Future**: sparse-view refinement, multi-frame mask re-initialisation, continuous learning for incremental teleconsultation updates.

Literature

Meuleman, A., Shah, I., Lanvin, A., Kerbl, B., & Drettakis, G. (2025). On-the-fly reconstruction for large-scale novel view synthesis from unposed images. *ACM Transactions on Graphics (TOG)*, 44(4), 1-14.
Ye, M., Danelljan, M., Yu, F., & Ke, L. (2024, September). Gaussian grouping: Segment and edit anything in 3d scenes. In *European conference on computer vision* (pp. 162-179). Cham: Springer Nature Switzerland.
Hu, X., Wang, Y., Fan, L., Luo, C., Fan, J., Lei, Z., ... & Zhang, Z. (2026). SAGD: Boundary-enhanced segment anything in 3D Gaussian via Gaussian decomposition. *IEEE Transactions on Image Processing*.
Schieber, H., Frischmann, D., Schaack, V., Boche, S., Schoellig, A., Leutenegger, S., & Roth, D. (2025). CoRe-GS: Coarse-to-Refined Gaussian Splatting with Semantic Object Focus. *arXiv preprint arXiv:2509.04859*.